

Research interests

Efficient LLM training, second-order and low-rank optimizers, FP4 quantized training, post-training quantization, Quantization Aware Training.

Education

- Oct 2024 – March 2025 **Research visit**, *UT Austin, VITA group*, Hosted by Professor Zhangyang "Atlas" Wang.
- 2021 – Dec 2026 **Computer Science PhD student**, *Aarhus University*, Denmark, Algorithms, Data and Artificial Intelligence group, expected graduation: December 2026.
- 2021–2024 **Masters in Computer Science**, *Aarhus University*, Denmark, Specialized in Algorithms and Machine Learning.
- 2018–2021 **Bachelor in Computer Science**, *Aarhus University*, Denmark.

Achievements

- 2022–2024 **Best paper award and Two Conference Oral presentations at International Conferences.**
- Received the best paper award for our work on quantized pre-training without storing high-precision weights [2] at WANT@ICML
 - Oral presentations: [2][4]
- 2020 **First Author Conference Paper Second Year at University**, *Published and presented a peer-reviewed paper at European Chapter of the ACL*[5].

Work Experience

- 2019–2021 **Student Programmer**, DANSKE BANK, Aarhus.
Worked on security systems: fraud detection, confidentiality and authentication solutions.
- 2021–2021 **Junior Trading Analyst**, DANSKE COMMODITIES, Aarhus.
Data analysis for commodities trading using machine learning techniques
- 2021–2025 **Teaching Assistant**, *Aarhus University*, Machine Learning Course.
- 2026–now **Research Intern**, *Qualcomm AI Research*, Amsterdam.
Working on improved methods for weight+Activation quantization

Publications

- Xiaoyuan Liang, S. Loeschcke, **Mads Tofttrup**, Anima Anandkumar. *M+Adam: Low-Precision Training via Additive-Multiplicative Optimization*. In *International Conference on Machine Learning (ICML)*, 2026.
Result: Alternative to AdamW for pretraining that does not require FP32 master weights
- S. Loeschcke*, **Mads Tofttrup***, M. J. Kastoryano, S. Belongie, and V. Snæbjarnarson. *LoQT: Low-Rank Adapters for Quantized Training*. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
Result: Enabled pretraining of LLM using 59% less memory compared to full-precision baseline with no quality drop
- † M. S. Bucarelli, M. F. Larsen, C. Schwiegelshohn, and **Mads Tofttrup**. *On Generalization Bounds for Projective Clustering*. In *Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
- † V. Braverman, V. Cohen-Addad, S. H. Jiang, R. Krauthgamer, C. Schwiegelshohn, **Mads Tofttrup**, and X. Wu. The Power of Uniform Sampling for Coresets. *Foundations of Computer Science (FOCS)*, 2022.
- Mads Tofttrup**, S. A. Sørensen, M. R. Ciosici, and I. Assent. *A reproduction of Apple's bi-directional LSTM models for language identification in short strings*. European Chapter of the ACL student workshop, (EACL), 2021.

* denotes equal contribution/joint first authorship. † denotes alphabetical author order, as is customary for works in Theoretical CS.